

CSIT545-02 - Computer Architecture Syllabus

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1. Textbook

We use the zyBook's *online textbook* of ***Computer Organization and Design: The Hardware/Software Interface, Fifth Edition***, by David Patterson and John Hennessy.

The zyBook contains the whole book (online access), and the accompanying exercises (or “activities”, also online access). Exercises will be assigned weekly as homework. All exercises will be recorded and scored, and will be part of your final grade. So it is **mandatory** that every student register a zyBook account to subscribe this e-textbook. Before the class begins, subscribe as follows:

1. Go to <https://learn.zybooks.com>
2. For the first time, go to upper-right to create an account **using your MSU email address**; once you've got an account, sign in
3. Enter zyBook code: **MONTCLAIRCSIT545WangFall2025**
4. Subscribe this textbook

The e-textbook+exercises will cost you \$97. Your homework assignments on Canvas are linked to zyBooks. They will be auto-graded by zyBooks and recorded to Canvas.

2. Subjects to be Covered

Chapters 2 through 6 of the textbook:

- Prerequisites: high-level language programming, assembly programming, fundamentals of digital design.
- The instruction set of MIPS computer.
- Arithmetic for computers: addition, subtraction, multiplication and division.
- Design of MIPS processor. Combinational approach and microprogramming technique.
- Pipeline technique for improving performance.
- Exploiting memory hierarchy: the concepts of caches, virtual memory and issues about them.
- Issues on processors-peripheral interfacing.
- Topics in parallel processing, multicores, multiprocessors, clusters.

3. Class Schedule

Class meets on Monday @5:20-8:05pm, in Room CCIS 152. Attendance to all classes is required, and will be recorded.

4. Office Hours

Monday @2:00-3:30PM. Other times also available by appointment.

5. Coursework, Exams, and Grades

As we cover subject materials, corresponding course activities will be assigned as homework every week. The purpose of these activities is to help reinforce the understanding of the materials discussed in class. All activities are available online only, via your zyBook account. Completed activities will be scored, and will be part of your final grade.

Homework activities must be completed on time. Treat homework due dates very seriously, and start homework early. **It is a firm policy that no late submissions will be graded, and no make-up for late submissions will be granted for ANY reasons.**

Same policy holds for midterm/final exams—**No make-up exams will be given for ANY reasons.**

The final grade is determined with the following distribution:

- Activities (homework) scores 30%
- Midterm exam 30%
- Final exam 40%